



REVIEW

Interleukin-4 and Atopic Dermatitis: Why Does it Matter? A Narrative Review

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ABSTRACT

Atopic dermatitis (AD) is a common chronic inflammatory skin condition that significantly impairs patients' quality of life as a result of intense itching and persistent eczematous lesions. Although AD has a multifaceted etiology—including genetic predisposition, environmental triggers, barrier dysfunction, and dysregulated immune responses—interleukin-4 (IL-4) has a recognized central role in its pathogenesis. This narrative review explores the role of IL-4 in

the pathophysiology of AD, its contribution to the atopic march, and the therapeutic impact of IL-4 inhibition. IL-4 plays a critical role in skin barrier dysfunction, dysbiosis, pruritus, and inflammation, all of which contribute to the debilitating symptoms of AD. Moreover, IL-4 is implicated in other atopic conditions, such as asthma, allergic rhinitis, and food allergies, underscoring its role beyond AD and its importance in the atopic march. Recent advances in targeted therapies, particularly IL-4/IL-13 signaling inhibitors, have changed AD management. Dupilumab, an IL-4 receptor antagonist, has

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demonstrated significant efficacy in reducing AD symptoms and enhancing patient outcomes in both children and adults. In addition to symptomatic relief, suppressing IL-4 signaling may also offer potential for disease modification, altering AD's progression and possibly preventing the onset of other atopic conditions. This review highlights the crucial role of IL-4 as a therapeutic target in AD. By understanding the role of IL-4 in AD pathogenesis and exploring the therapeutic implications of targeting IL-4 pathways, this work can contribute to guide future research concerning treatment approaches and also emphasize the need for early and targeted interventions to mitigate disease impact and ultimately improve patient quality of life.

Keywords: Atopic dermatitis; Interleukin-4; Targeted therapies; Atopic march; Disease modification; Pruritus; Skin barrier dysfunction; Inflammation; Dysbiosis; Th2 inflammation

Key Summary Points

Interleukin-4 (IL-4) plays a crucial role in the pathogenesis of atopic dermatitis (AD), as it contributes to skin barrier dysfunction, pruritus, dysbiosis, and inflammation, which are the four cardinal pillars of the condition.

IL-4 is implicated in the pathogenesis of other atopic conditions such as asthma, allergic rhinitis, and food allergies, making it a key player in the atopic march and underscoring its importance beyond AD.

Targeting IL-4/IL-13 signaling, particularly with targeted therapies like dupilumab, has revolutionized AD management, leading to significant improvements in patient symptoms, quality of life, and overall disease control.

Dupilumab has shown some potential to impact disease modification, with the possibility of altering AD progression and preventing the development of other allergic diseases in certain populations when introduced early.

INTRODUCTION

Atopic dermatitis (AD), also known as atopic eczema, is a chronic inflammatory skin disease that causes intense itching and recurrent eczematous lesions [1]. Characterized by debilitating signs and symptoms—intense itching (pruritus), eczematous lesions, and impaired skin barrier—AD disrupts physical health and significantly impacts patients' emotional and psychological well-being, often requiring long-term management and affecting overall quality of life [2]. Globally, AD prevalence has been increasing, affecting 3–10% of adults and 10–20% of children [3], with notable regional variations. This underscores the need for a better understanding of the underlying mechanisms of AD and the development of more effective treatment strategies [4–6].

The precise etiology of AD is unclear, but a growing body of evidence implicates a complex interplay of genetic predisposition, environmental triggers, epidermal barrier dysfunction, dysbiosis, and dysregulated immune responses [7]. Central to this process is the role of interleukin-4 (IL-4), a cytokine that drives T helper 2 (Th2) cell-mediated pathways, exacerbating inflammation, barrier disruption, and other characteristics of AD [8]. Beyond its direct role in AD, the effects of IL-4 extend to other atopic conditions such as asthma and allergic rhinitis, highlighting its broader significance in atopic comorbidities [9–11].

In recent years, significant progress has been made in understanding the therapeutic implications of IL-4 inhibition. Clinical and preclinical studies have shown promising outcomes, suggesting that targeting IL-4 can lead to substantial improvements in disease signs and symptoms and overall quality of life for patients with AD [12, 13]. This comprehensive review aims to explore the multifaceted role of IL-4 in AD, focusing on its involvement in several mechanisms known to be hallmarks of AD, namely epidermal barrier disruption, chronic inflammation, pruritus, dysbiosis, and immunoglobulin E (IgE) production. Additionally, the impact of IL-4 inhibition will be discussed, including its potential for disease modification and its

significance in the context of other therapeutic approaches. By understanding the critical role of IL-4, this review seeks to understand its role in AD and how targeting the IL-4 pathways can enlighten further treatment approaches.

Ethical Approval

This study did not involve new studies with human participants or animals. The Open Access Fees of the publication were sponsored by Sanofi Portugal.

PATHOPHYSIOLOGY OF ATOPIC DERMATITIS

The pathogenesis of AD involves a complex interplay of genetic susceptibility, epidermal barrier disruption, immune dysregulation, and skin dysbiosis [6, 14, 15]. Skin barrier dysfunction has been traditionally considered the first step for the onset of AD and the subsequent development of the atopic march [16]. Some patients with AD have an underlying genetic predisposition, mainly mutations in the filaggrin (*FLG*) gene, which encodes a critical structural protein in the stratum corneum involved in maintaining skin barrier integrity. *FLG* mutations, particularly homozygous, are associated with an increased risk of severe AD with earlier onset, longer persistence, and skin infection, and they occur in 50% of children with moderate to severe AD [17–19]. Filaggrin deficiency leads to impaired skin barrier function, characterized by increased transepidermal water loss and decreased structural stability of the skin. This compromised barrier facilitates the penetration of irritants, allergens, and colonization by microbes, particularly *Staphylococcus aureus* [15, 20–22].

While *FLG* mutations have been the most studied, other epidermal barrier genes—loricrin (*LOR*), filaggrin-2 (*FLG-2*), hornerin (*HRNR*), *SPINK5*, *SPRR3*, desmoglein 1 (*DSG1*), involucrin (*IVL*), and claudin-1 (*CLDN1*)—as well as immune system-related genes—Toll-like receptor 2 (*TLR2*), *TLR4*, *TLR9*, nucleotide-binding oligomerization domain-containing protein 1

(*NOD1*), *NOD2*, and interferon-gamma (*IFN γ*)—have been associated with AD. The prevalence of these genetic variants differs across populations, highlighting the complex and heterogeneous nature of AD [19, 23].

Upon disruption of the skin barrier, keratinocytes, the predominant cells in the epidermis, release various immune-regulating molecules called alarmins, such as IL-1 β (IL-1 β), IL-25, IL-33, and thymic stromal lymphopoietin (TSLP). These molecules play a crucial role in activating the immune system, particularly Th2, Th17, and Th22 immune responses [15, 16]. Th2 cells release cytokines like IL-4 and IL-13, which further weaken the skin barrier by downregulating the expression of essential proteins like filaggrin. Additionally, TSLP, IL-25 and IL-33 promote innate lymphoid cells type 2 (ILC2) expansion, which leads to the production of IL-4, IL-13, IL-9, and IL-5, further impairing skin barrier [24]. IL-4, IL-13, and IL-31 also affect keratinocytes' differentiation and trigger increased pruritus [25]. This creates a vicious cycle of barrier dysfunction and inflammation, perpetuating the disease state [21, 22]. Indeed, the progression from acute to chronic AD involves not only a quantitative intensification of the Th2, Th22, and Th17 responses but also a notable upregulation of Th1-associated genes, resulting in increased secretion of IFN γ and IL-2 [26, 27]. However, the relative importance of these subtypes of T cells can vary according to ethnicity [28].

The itch-scratch cycle is another key component of AD pathophysiology. The release of Th2 cytokines such as TSLP, IL-2, IL-4, IL-13, and IL-31 combined with amplified sensory nerve signals intensifies itching [1, 21, 29]. Indeed, IL-31 has been recently identified as a key player in pruritus, along with IL-4 that can signal directly through receptors expressed on various branches of epidermal and dermal nerve endings [21]. Contrary to earlier assumptions, histaminergic pathways play a less significant role in AD, with histamine from mast cells contributing minimally to the itch-scratch cycle. The sensation of itch induces scratching, which further damages the skin barrier, leading to more inflammation and a heightened immune response.

Skin dysbiosis is an important hallmark of AD [1]. As a result of impaired epidermal barrier function and dysregulated antimicrobial immunity, the skin of patients with AD is highly susceptible to bacterial infections, particularly from *S. aureus* [6, 30]. The skin microbiome of patients with AD is marked by reduced diversity of bacterial strains and increased colonization of *S. aureus*, which is present on the skin of 60–100% of those affected, comparing with 5–30% of healthy subjects [6]. *S. aureus* contributes to AD pathogenesis in several ways, including further barrier disruption with the production of ceramidases and direct proinflammatory effects such as type 2 immune responses [1]. Additionally, other coagulase-negative staphylococcus strains, *Staphylococcus epidermidis* and *Staphylococcus hominis*, are increased in AD skin [30].

Recently, gut dysbiosis has also been linked to AD pathogenesis [31]. It results from an imbalance in gut microbiota that leads to immune dysregulation and barrier disruption, allowing inflammatory substances to permeate the bloodstream and reach the skin. This exacerbates the Th2 immune response and contributes to tissue damage and sensitization to these chemicals. Moreover, dietary antigens, bacterial toxins, and pathogens may accumulate in the skin, further impairing skin barrier and perpetuating inflammation [31–33]. This systemic aspect of AD illustrates the interconnectedness of different body systems in the disease pathogenesis [34, 35].

THE ROLE OF IL-4 IN THE FOUR CARDINAL PILLARS OF ATOPIC DERMATITIS

IL-4 plays a pivotal role in the pathophysiology of AD, influencing each of the four cardinal pillars of AD: pruritus, inflammation, skin barrier dysfunction, and dysbiosis (Fig. 1).

IL-4 signals through two different receptor complexes: a type I receptor composed of IL-4R α chain and common γ chain (IL-4 R γ subunit), and a type II receptor consisting of IL-4R α and IL-13R α 1 subunits. The type I receptor specifically transduces signals for IL-4 and it

is primarily expressed on hematopoietic cells, such as T cells. In contrast, the type II receptor can be activated by either IL-4 or IL-13. This receptor is expressed on a broader range of cells, including non-hematopoietic cells such as fibroblasts, keratinocytes, and other epithelial cells, which are key players in tissue inflammation and remodeling. While both IL-4 and IL-13 can signal through the type II receptor, the effects of IL-13 are restricted to cells expressing this receptor, which explains why IL-13 may not induce Th2 differentiation directly—as T cells express the type I receptor but not the type II receptor [36–38].

In the development of allergic inflammation, IL-4 and IL-13 have overlapping and additive roles [38]. IL-4 promotes the development of Th2 cells, while IL-13 is critical in promoting tissue inflammation, acting on non-hematopoietic cells through the type II receptor. IL-4 is involved in Th2 differentiation and controls lymphocyte functions such as IgE synthesis in B cells [39, 40]. Hence, IL-4 has a more central action in AD, as it acts in the lymph nodes, whereas IL-13 plays a more peripheral role, exerting its effects mainly in the skin. Th2 cells, ILC2, mast cells, eosinophils, and basophils all produce both IL-4 and IL-13 [38, 40].

Skin Barrier Dysfunction

IL-4 significantly impairs skin barrier function, a hallmark of AD. IL-4 stimulation in vitro changes early and terminal keratinocytes differentiation, suppressing the expression of both terminal keratinocyte differentiation proteins (i.e., filaggrin, loricrin, and involucrin) and genes related to differentiating keratinocytes at early stages. The downregulated proteins are critical for maintaining the integrity of the skin barrier, leading to the disruption of tight junctions and impaired ceramide production in the skin [39, 41–43]. Consequently, this causes increased transepidermal water loss, which results in a compromised barrier that is more susceptible to irritants, allergens, and microbial invasion [15, 22, 42, 43].

Furthermore, IL-4 alters lipid metabolism and content in the skin by reducing the production

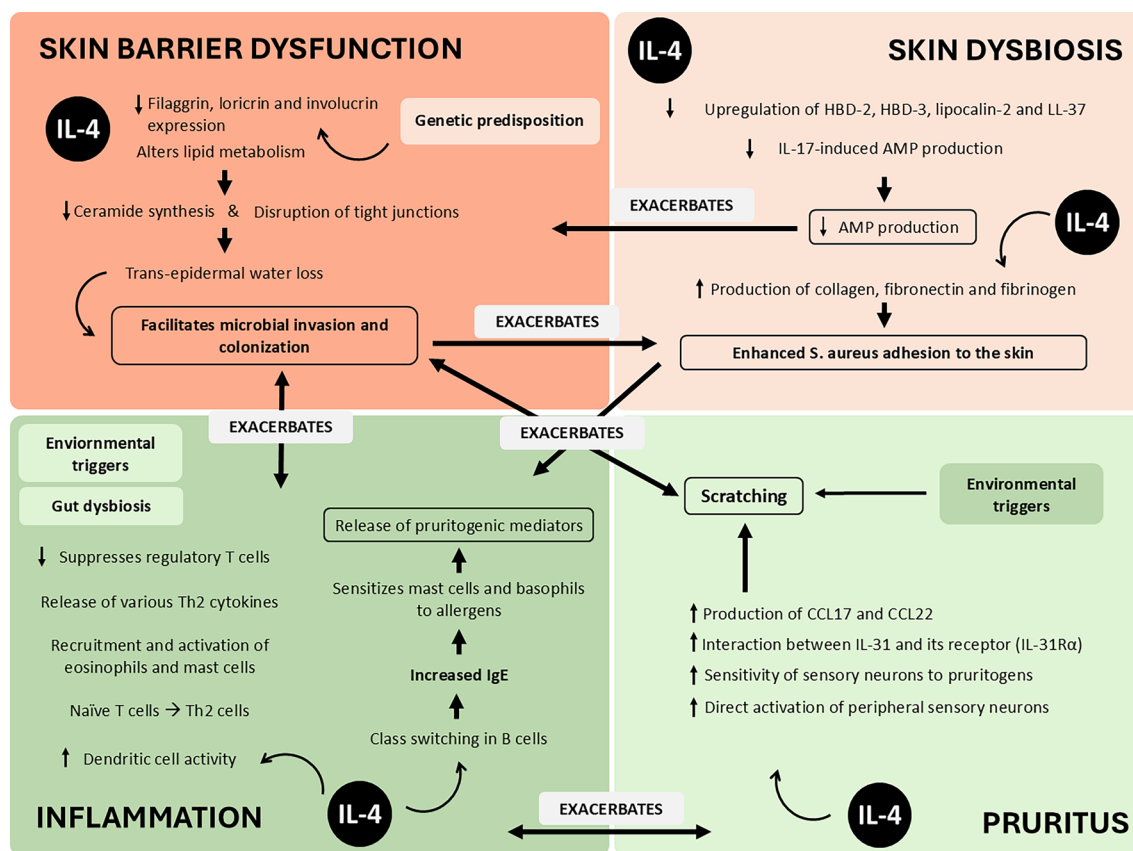


Fig. 1 The role of IL-4 in the four cardinal pillars of atopic dermatitis: skin barrier dysfunction, skin dysbiosis, inflammation, and pruritus

of lipids like ceramides crucial for barrier function. Patients with AD exhibit reduced levels of total ceramide and free fatty acids [44, 45]. In particular, larger species of ceramides (>40 total carbons, like Cer[NH], Cer[EOS], Cer[NP], Cer[EOH], and Cer[EOP]), which are essential for the formation of a tightly packed lipid barrier are expressed at significantly lower levels, whereas smaller ceramide species (<40 total carbons), that render the skin more permeable, are expressed at higher levels [45]. IL-4 and IL-13 were shown to downregulate the expression of lipid chain elongases 3 and 6 (ELOVL3, ELOVL6) in keratinocytes, highlighting the crosstalk between inflammatory response and skin barrier components [45]. In addition, the deficient lipid composition that characterizes patients with AD correlates with altered expression of other enzymes also involved in ceramide and fatty acids biosynthesis, like β -glucocerebrosidase

(GBA), acid-sphingomyelinase (aSmase), ceramide synthase 3 (CerS3), stearyl CoA desaturase (SCD), and ELOVL1 [46]. This lipid deficiency further exacerbates barrier dysfunction and facilitates the chronic inflammatory state observed in AD [42, 47].

Skin Dysbiosis

Concerning the skin microbiome, IL-4 has been linked to increased *S. aureus* adhesion and penetration into the epidermis, which correlates not only with elevated levels of IL-4 but also other cytokines (IL-13, IL-22, and TSLP), along with decreased expression of antimicrobial peptides (AMPs), critical for maintaining skin integrity and microbial balance [48]. Specifically, IL-4 attenuates AMP production by suppressing upregulation of beta defensins

(HBD-2 and HBD-3), lipocalin-2, and LL-37, even antagonizing IL-17-induced AMP production in keratinocytes, which further compromises the skin's defensive capabilities [41].

Moreover, IL-4 promotes fibroblast production of extracellular matrix components like collagen, fibronectin, and fibrinogen, which enhance *S. aureus* adhesion to the skin [41]. This interaction underscores the contribution of IL-4 to both skin barrier dysfunction and increased susceptibility to bacterial infections. Consequently, this dysregulation contributes to an environment that is not only more susceptible to bacterial infections but also exacerbates the overall inflammatory response associated with AD.

Pruritus

IL-4 is paramount for chronic itch in AD through multiple direct and indirect mechanisms. By acting on the IL-4R α expressed on the surface of the peripheral sensory neurons, IL-4 promotes their direct activation, releasing different pruritogenic mediators and increasing their responsiveness to subthreshold levels of several pruritogens [41, 49–51].

Furthermore, IL-4 amplifies the role of IL-31 in pruritus by enhancing the interaction between this critical cytokine and its receptor (IL-31R α). This interaction leads to increased production of CCL17 and CCL22. IL-4 also upregulates the expression of IL-31R α on sensory neurons, further exacerbating the itch response. This persistent itching not only leads to further skin damage through scratching but also significantly impacts the quality of life of patients with AD [52].

Inflammation

IL-4 is a central player in both the chronic and acute inflammatory processes of AD. IL-4 reduces dendritic cells' expression of IL-12, MHC class II, and costimulatory molecules, while increasing IL-10 production. This favors the differentiation of naïve CD4⁺ and CD8⁺ T cells toward type 2 cells. Additionally, IL-4

stimulates the expression of chemokine C–C motif ligand (CCL) CCL26 in keratinocytes and CCL11 in fibroblasts, which are critical for eosinophil recruitment [41, 53]. The recruitment of eosinophils, along with other inflammatory cells like mast cells, causes the release of pro-inflammatory mediators that sustain chronic inflammation in the skin. In addition, IL-4 supports the development of ILC2 cells, creating a positive feedback loop that enhances the Th2 response [16, 41, 43]. IL-4 also inhibits the function of regulatory T cells (Tregs), which are crucial for maintaining immune tolerance and regulating inflammatory responses [14, 54].

IL-4 is a key driver of IgE production, a relevant component of the AD response [55]. It promotes class switching in B cells, leading to the production of IgE antibodies. These antibodies bind to mast cells and basophils, sensitizing them to allergens. Upon exposure to allergens, these sensitized cells release inflammatory mediators, contributing to the inflammation characteristic of AD [41, 56, 57]. In cases of IgE-allergic AD, the binding of IgE to receptors on dendritic cells, like Langerhans cells and inflammatory dendritic epidermal cells, enhances allergen uptake and initiates hypersensitivity reactions [58]. The elevated levels of IgE not only perpetuate the allergic response but also exacerbate skin inflammation and barrier dysfunction, creating a feedback loop that sustains the disease [14, 41]. In fact, the reduction in IgE levels during treatment with dupilumab appears to correlate with a decrease in AD flares, emphasizing its key role in the disease's pathogenesis [59].

Table 1 highlights the multifaceted role of IL-4 in AD.

THE EXPANSIVE ROLE OF IL-4 IN THE ATOPIC MARCH

AD is often associated with other atopic conditions, such as asthma, allergic rhinitis, and food allergies, known as atopic comorbidities [62]. The term “atopic march” refers to the typical progression of these allergic diseases,

Table 1 Multifaceted role of IL-4 in atopic dermatitis

Pathophysiological aspect	Role of IL-4	Impact	References
Compromised skin barrier	Downregulates filaggrin, loricrin, and involucrin expression	Disruption of tight junctions and impaired ceramide production	Upadhyay et al. 2023 [42]; Clausen et al. 2015 [22]
	Alters lipid metabolism in the skin	Reduced production of barrier lipids and further barrier dysfunction	Upadhyay et al. 2023 [42]
	Decreases ceramide synthesis in the stratum corneum	Weakened skin barrier and increased transepidermal water loss	Sawada et al. 2012 [60]
Skin dysbiosis	Facilitates microbial invasion and colonization	Increased susceptibility to <i>S. aureus</i>	Bao et al. 2012 [43]
	Decrease in HBD-2, HBD-3, lipocalin-2, and LL-37	Diminishes AMP production further compromising skin barrier	Chiricozzi et al. 2020 [41]; Nakatsuji et al. 2016 [48]
	Antagonizes IL-17-induced AMP production in keratinocytes		
	Increased production of extracellular matrix components (collagen, fibronectin, and fibrinogen)	Enhanced <i>S. aureus</i> adhesion to the skin	Chiricozzi et al. 2020 [41]
Pruritus	Activates sensory neurons and induces keratinocytes	Severe itching and activation of pruritogenic pathways	Mollanazar et al. 2016 [29]; Kido-Nakahara et al. 2016 [52]; Oetjen et al. 2017 [50]
	Induces peripheral sensitization to subthreshold levels of pruritogens	Exacerbation of the itch–scratch cycle	
	Augments the interaction between IL-31 and its receptor	IL-31 upregulates CCL17 and CCL22 production, intensifying pruritus	Miake et al. 2019 [49]; Kido-Nakahara et al. 2017 [52]
Inflammation	Promotes differentiation of naïve T cells into Th2 cells	Amplification of Th2 immune response	Dubin et al. [12]; Chiricozzi et al. 2020 [41]
	Recruits and activates eosinophils and mast cells	Sustained release of pro-inflammatory mediators	Bao et al. 2012 [43]
	Enhances dendritic cell activity	Increased antigen presentation and Th2 activation	Teraki et al. 2000 [61]
	Suppresses regulatory T cell (Treg) function	Unchecked inflammatory response	Radi et al. 2022 [14]; Cui et al. 2021 [54]

Table 1 continued

Pathophysiological aspect	Role of IL-4	Impact	References
Increased IgE production	Promotes class switching in B cells to IgE production	Elevated IgE levels and allergic sensitization	Chiricozzi et al. 2020 [41]
	Sensitizes dendritic cells, mast cells, and basophils to allergens	Enhanced allergic reactions and pruritogen release	Gauchat et al. 1993 [56]; Niggemann et al. 1997 [57]
Other factors	Induces production of other Th2 cytokines (e.g., IL-13 and IL-5, IL-31)	Synergistic effects on inflammation and barrier dysfunction	Yamamura et al. 2024 [40]
	Affects gut–skin axis by promoting gut dysbiosis	Systemic inflammation and exacerbation of AD symptoms	Lee et al. 2018 [34]

often beginning with AD and progressing sequentially, while subsequently increasing the likelihood of developing the others and amplifying the allergic burden [63]. While the concept of atopic march is well established, recent research suggests that this progression does not follow a linear path for all individuals, giving rise to the idea of “atopic trajectories” to describe the various pathways through which allergic conditions can develop over time [64, 65].

IL-4 plays a central role in the pathogenesis of the atopic comorbidities by promoting allergic inflammation and IgE production, whose contribution is more important in these comorbidities. The involvement of IL-4 in these conditions underscores its expansive role in the atopic march, highlighting the need for targeted therapies that address IL-4-mediated pathways across different atopic diseases [66].

Asthma

Approximately one-third of patients with AD develop asthma. IL-4 plays a significant role in asthma, as evidenced by the increase of IL-4Ra expression in bronchial biopsy specimens from atopic and non-atopic asthmatic subjects

[67]. Besides, IL-4 induces IgE isotype switch, increases expression of vascular cell adhesion molecule 1 (VCAM-1), promotes eosinophil transmigration, stimulates mucus production, and aids in the differentiation of Th2 lymphocytes, leading to cytokine release [68]. While its role often overlaps with IL-13, IL-4 also acts independently. Evidence suggests that IL-4 exerts a superior influence, since the inhibition of both cytokines with dupilumab has shown benefits in moderate to severe asthma—by reducing exacerbations, and symptoms and improving quality of life—while single inhibition of IL-13 by lebrikizumab and tralokinumab did not induce significant improvements [69, 70].

Rhinitis

Many patients with AD frequently suffer from allergic and non-allergic rhinitis. IL-4 is overexpressed in allergic rhinitis patients, which drives the allergic response, particularly through IgE production [9, 71]. Recent data indicated that inhibition of IL-4 and IL-13 with dupilumab reduced symptom burden in allergic rhinitis, particularly by suppression of IgE production [72]. In contrast, targeting only IL-13 has not reached the expected results [9].

Food allergy

A notable proportion of adult patients with AD have food sensitivity (28.6%) and food allergy (24.1%), with prevalence rates even higher in children [62]. A recent meta-analysis described a pooled prevalence of 40.7% for challenged-proven food allergy in patients, ranging from 34.1% to 47.5% [73]. IL-4 is implicated in this condition by regulating Th2 cell proliferation and survival and IgE synthesis, leading to disruption of the gut barrier and increased permeability of these allergens. Recent findings showed that dupilumab induced a significant decrease in food-specific IgE levels in children with AD, reinforcing the role of the IL-4/IL-13 axis in food allergies [74].

TARGETING IL-4: CLINICAL OUTCOMES AND IMPLICATIONS

Inhibition of IL-4: Lessons from In vitro studies and Animal Models

The inhibition of IL-4R α has been a focal point of therapeutic research in allergic diseases, including AD. Several in vitro and in vivo studies show the potential of IL-4R antagonism in modulating allergic responses central to these conditions [51].

In vitro studies showed that IL-4R blocking reduces IL-4-activities, such as B cell and T cell proliferation and differentiation into Th2 cells [75, 76]. Consequently, it decreases B cell activities like IgG1 and IgE secretion [77]. Moreover, suppression of IgE production was corroborated by a study with an IL-4R antagonist which inhibited IL-4- and IL-13-induced STAT6 phosphorylation [78].

Inhibiting IL-4R in animal models has demonstrated considerable effects in mitigating allergic symptoms. Early research revealed that soluble IL-4R blocked IgE secretion up to 85% by neutralizing endogenous IL-4. This finding highlighted the receptor's potential not only as an antagonist but also as modulator of cytokine delivery in therapeutic contexts [79]. In mice

models of asthma—characterized by irregular tissue structure and severe inflammation—blocking IL-4R resulted in more regular tissue structure and light inflammation, due to inhibited IL-4/STAT6 pathway [80]. Furthermore, IL-4R α antagonists were shown to inhibit allergen-induced airway eosinophilia and hyperresponsiveness in murine models [78].

Altogether, these studies highlighted the potential of IL-4 signaling suppression in treating atopic conditions. Preclinical models demonstrated that IL-4R α antagonism effectively reduces inflammation and IgE production, laying the foundation for biologic therapies targeting diseases like AD.

The Rise of Targeted Therapies

The rise of targeted therapies—biologic and non-biologic agents—marks a significant shift in the management of AD and other allergic disorders, offering hope for long-term disease control and better patient outcomes [81]. Historically, AD treatment has relied in topical and oral corticosteroids and other immunosuppressive medications. However, with the increasing understanding of the complex pathophysiology of AD, more precise targeted therapies have been developed [82, 83].

Among the less targeted therapies are janus kinase (JAK) inhibitors. These small molecules target multiple pathways involved in AD by blocking the activity of the JAK family of enzymes (JAK1, JAK2, JAK3, and TYK2), subsequently disrupting cytokine signaling, particularly IL-4, IL-13, and IL-31 [81]. This broader inhibition allows JAK inhibitors to modulate a wide range of immune responses; however, their less selective nature compared to biologics also increases the risk of off-target effects and associated side effects [84, 85]. Nonetheless, JAK inhibitors like abrocitinib, upadacitinib, and baricitinib have shown efficacy and tolerability, making them a valuable option for certain patient populations [86].

In contrast, biologic therapies directed at the IL-4/IL-13 pathway offer a more targeted approach to AD management, by inhibiting two of the key cytokines to the disease process. As a

result of the established importance of IL-4 in AD, therapies that block IL-4R α have attracted significant interest. The breakthrough came with dupilumab, the first biologic approved in 2017 for moderate to severe AD, which significantly improved treatment outcomes by inhibiting both IL-4 and IL-13 signaling through their shared receptor (IL-4R α).

Dupilumab, a fully human IgG4 monoclonal antibody (mAb), has demonstrated significant improvements in disease outcomes in clinical trials, reducing symptoms such as pruritus, anxiety, and depression, while also improving skin clearance and overall quality of life. These benefits have been sustained over time, even in real-world settings, with a favorable safety profile [87–89]. The most frequent side effects were ocular conditions, particularly conjunctivitis, observed in up to 28% of dupilumab users in clinical trials and up to 38.2% in real-world settings [14]. Other reported adverse events in patients treated with dupilumab were nasopharyngitis, headache, injection site reaction, and an increase in oral herpes infections [90].

Dupilumab has also been evaluated in children and adolescents, and the efficacy and safety appear to be similar [14, 91]. Trials in children aged 6 months to 6 years and 6 to 12 years with severe AD reported significant improvements, with mostly transient mild to moderate adverse events [92, 93]. A phase 3 trial in the same age group also reported significant improvements in symptoms and quality of life, although dupilumab was administered alongside topical corticosteroids [94].

The success of dupilumab has sparked the development of other monoclonal antibodies targeting the IL-4/IL-13 axis. Rademikibart (CBP-201), directed at IL-4R α , has shown promise in clinical trials by suppressing Th2-driven inflammation, similarly to dupilumab, and significantly decreasing EASI and pruritus, with a favorable safety profile [95, 96]. Manfidokimab (AK120), also directed at IL-4R α , showed preliminary efficacy and an acceptable safety profile [97]. Another example is eblasakimab (ASLAN004), a first-in-class, fully human mAb that prevents IL-4/IL-13 axis signaling by binding to the IL-13R α 1 subunit of the type 2 receptors. This does not completely

inhibit IL-4 signaling, resulting in a partial modulation of IL-4 effects, while still significantly impacting the overall inflammatory response [98].

Therapies selectively targeting IL-13 and without interfering with IL-4 signaling have also been developed. Lebrikizumab, recently approved for use in patients over 12 years old, impairs the heterodimerization of the IL-4R α –IL-13R α 1 signaling complex, thus blocking only IL-13 signaling. Phase 3 trials reported improved skin clearance, itch, and sleep in patients, with a favorable safety profile [99, 100]. Additionally, tralokinumab, a fully human mAb which binds to IL-13 and blocks its interaction with the receptor, has shown significant improvements in AD compared with placebo, but it was associated with more injection site reactions, compared to dupilumab [81, 101, 102]. While these IL-13-targeting therapies have demonstrated improvements in AD signs and symptoms, they were also associated with some adverse events, including conjunctivitis and injection site reactions. Although these therapies are beneficial, they may not fully address the broader immune dysregulation seen in AD compared to IL-4/IL-13 inhibition.

Treating AD by targeting either the IL-4/IL-13 axis or IL-13 alone has proven effective, but the central role of IL-4 in AD cannot be overlooked. IL-4 is critical in Th2 cell differentiation, IgE class switching, and the promotion of chronic pruritus makes it a more comprehensive target for addressing the immune dysregulation underlining AD. Unlike IL-13, which predominantly acts at the peripheral level, particularly in the skin, IL-4 is essential for driving systemic immune responses, making it a central player in both the immune system's central processes and peripheral tissue inflammation. Table 2 shows an overview of the differential roles of these two cytokines in AD.

Table 2 Differential role of IL-4 and IL-13 in the pathophysiology of atopic dermatitis

	IL-4	IL-13
Skin barrier dysfunction and dysbiosis	Downregulates key structural proteins and lipids Decreases AMP production Increased production of extracellular matrix components (collagen, fibronectin, and fibrinogen)	Downregulates key structural proteins and lipids Activates STAT6, leading to <i>FLG</i> and <i>IVL</i> suppression Stimulates periostin and IL-24 pathways
Pruritus	Activates sensory neurons and induces peripheral sensitization to pruritogens Augments the interaction between IL-31 and its receptor	Promotes itch via the TRPA1 pathway in sensory neurons Induces histamine-independent itch and amplifies itch responses
Inflammation	Plays a central role by promoting differentiation of naïve T cells into Th2 cells Recruits and activates eosinophils and mast cells Enhances dendritic cell activity Suppresses regulatory T cell (Treg) function Promotes class switching in B cells, increasing production of IgE antibodies	Has a peripheral role by amplifying Th2-driven inflammation through the JAK-STAT pathway (specifically STAT3, STAT1, STAT6) Promotes eosinophil recruitment and the release of pro-inflammatory cytokines Indirectly supports IgE production through Th2 cell activation and cytokine release
Atopic march	Has a prominent role in the atopic comorbidities, for instance asthma, allergic rhinitis, and food allergy/sensitivity	Apparently not so important in the atopic comorbidities, but may exacerbate chronic skin inflammation and itch

Can IL-4 Inhibition Promote Disease Modification?

Disease modification in AD refers to interventions that can alter the natural course of the disease, potentially preventing or delaying the progression to more severe forms and associated comorbidities [103]. This is particularly relevant for the atopic march, the progression from AD in early childhood to the development of other atopic conditions like asthma, allergic rhinitis, and food allergies in adulthood [104].

In AD, the atopic march or atopic trajectories only occur in a subset of patients—approximately two-thirds—with the risk being higher in children who produce IgE antibodies in response to environmental triggers [63, 105]. By blocking both IL-4 and IL-13 signaling, dupilumab has shown significant potential for disease modification in AD. Treatment with dupilumab has demonstrated improvements in disease severity, while also reducing serum total

IgE levels in adolescents and adults [63]. This suggests that by targeting the key cytokines driving the Th2-mediated inflammation in AD, it may be possible to slow or even prevent the progression to other atopic conditions. A meta-analysis based on data from clinical trials documented a 34% reduction in new allergies in patients over 12 years old, an effect that persisted even during off-treatment periods. This meta-analysis supported that dupilumab may have potential for disease modification, particularly in adolescents, patients with early disease onset and with high baseline IgE levels. It is still unknown if the effect is due to the combined inhibition of IL-4/IL-13 or to IL-4 or IL-13 alone, as there are no similar data available with selective IL-13 inhibitors. However, on the basis of the current knowledge of IL-4, its inhibition appears to be crucial for the pathogenesis of allergic comorbidities. Future studies are required to explore this topic further and also because real-world evidence is scarce, particularly in children [63].

A recent real-world study in pediatric patients demonstrated that those on dupilumab had a significantly decreased risk of developing atopic march towards asthma or allergic rhinitis (32%), compared with patients on conventional therapy [105]. This effect was more pronounced in younger children, emphasizing the need for early intervention to maximize long-term benefits [63, 105]. The best opportunity to intervene for disease modification and to avoid the development of the atopic comorbidities is probably during childhood, when the immune system and skin barrier are still developing. Indeed, beyond the window of opportunity, even with AD remission via dupilumab, the mechanisms driving atopic march may already be active [63, 105, 106].

The disease-modifying potential of IL-4/IL-13 inhibition, as seen with dupilumab, also introduces the possibility of remission of AD after treatment discontinuation. Remission refers to the sustained improvement or resolution of disease symptoms and signs, even after the discontinuation of the therapeutic intervention, which has been demonstrated for up to 40 weeks in adults [107]. An ongoing study presented at 2024 Society for Pediatric Dermatology Annual Meeting documented that approximately one-third of pediatric patients achieved clinical remission, and half of them maintained therapy-free remission [108]. Data from the Bioday registry also emphasizes the sustained effectiveness of dupilumab up to 5 years after discontinuation of the treatment [109]. More comprehensive data on how dupilumab maintains its effects for longer follow-up periods would further strengthen the case for its role in disease modification, which is still unclear. This could have profound implications for the management of AD, potentially allowing for more durable control of the condition and a reduced need for continuous treatment, particularly when intervention is made in childhood [106]. However, it should be noted that, particularly in childhood, spontaneous remission is a possibility and should always be considered, which adds complexity to the interpretation on off-therapy AD remission.

A question remains whether therapies targeting only IL-13, such as tralokinumab and

lebrikizumab, may also have potential disease-modifying ability. By targeting both cytokines, dupilumab has the potential to block the biological activity of IL-13 as the most important driving force of the skin inflammation, as well as the more centrally acting IL-4. This dual targeting may give dupilumab a greater impact on the overall disease, but additional data is warranted [110].

FUTURE PERSPECTIVES AND IMPACT ON CLINICAL PRACTICE

The future of AD treatment is set to face significant changes, as a result of the ongoing development of biologic therapies and small molecules that may provide more precise and effective interventions. The successful introduction of dupilumab, which targets the IL-4/IL-13 axis, has paved the way for a new era in AD management [63, 90, 111].

If future studies, particularly in pediatric populations, confirm dupilumab's ability to modify the course of AD and the atopic march, it could become the first disease-modifying treatment for AD. In addition, other emerging therapies, including OX40/OX40L inhibitors, which target earlier stages of the immunologic cascade, may also show disease-modifying potential. Furthermore, the role of broader-acting agents, like JAK inhibitors, in providing similar long-term benefits remains an area of investigation. As the field of advanced therapies expands, future management may focus on combining biologics with small molecules, bispecific antibodies, ensuring maximal efficacy with minimal adverse effects [81].

Integrating personalized medicine into AD management presents as an opportunity for more effective treatment strategies. Advances in genomic research and biomarker identification could enable treatments to be tailored according to individual patient profiles [111]. For example, predictive models could be used in future clinical practice to determine the optimum management alternatives for each individual [111, 112]. The implications for clinical practice will be profound, particularly in its potential to modify

the disease course rather than simply managing signs and symptoms.

CONCLUSION

There is robust evidence supporting the crucial role of IL-4 in the pathogenesis of AD, despite the challenges to distinguish its effect from IL-13. This underscores the strong rationale to target IL-4 in the management of AD. Inhibiting specifically IL-4/IL-13 signaling has led to significant improvements in symptoms, quality of life, and overall disease management, unveiling the potential for disease modification—a capability not yet proven. When introduced early in the course of the disease, dupilumab may not only alleviate current symptoms, but also prevent the onset of associated allergic diseases, positioning IL-4 as a central player in altering the disease trajectory.

As we deepen our understanding of this critical cytokine, the potential for novel therapies continues to expand. With ongoing research and innovative approaches, there is hope for more effective and comprehensive management strategies that could redefine AD treatment in the years to come. Such advancements could ultimately improve the quality of life for millions affected by this debilitating condition.

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Declarations

Conflict of Interest. Tiago Torres has received consultancy and/or speaker's honoraria from and/or participated in clinical trials sponsored by AbbVie, Amgen, Almirall, Amgen, Apogee Therapeutics, Arena Pharmaceuticals, Biocad, Biogen, Boehringer Ingelheim, Bristol Myers Squibb, Celgene, Fresenius-Kabi, Johnson & Johnson Innovative Medicine, LEO Pharma, Eli Lilly, MSD, Mylan, Novartis, Pfizer, Samsung-Bioepis, Sanofi-Genzyme, Sandoz, STADA and UCB. Tiago Torres is also an Editorial Board member of *Dermatology and Therapy*. Tiago Torres was not involved in the selection of peer reviewers for the manuscript nor any of the subsequent editorial decisions. Pedro Mendes-Bastos has received consultancy and/or speaker's honoraria from and/or participated in clinical trials sponsored by Lilly, AbbVie, Janssen, Novartis, LEO Pharma, Almirall, Sanofi, Regeneron, Viatris, Pierre Fabre, L'Oréal, Organon, Evelo Biosciences, CS Portugal, Pfizer, Biogen, Apogee, Amgen. Margarida Gonçalves has participated in clinical trials, advisory board and/or lectures promoted by Abbvie, Almirall, Amgen, Astra-Zeneca, Biogen, Eli Lilly, Janssen, Leo Pharma, Novartis, Pfizer, Sanofi-Genzyme, Takeda. Maria João Cruz has participated in clinical trials, advisory board and/or lectures promoted by Abbvie, Almirall, Amgen, Janssen, Leo Pharma, Eli Lilly, Novartis, Pfizer and Sanofi-Genzyme. Bruno Duarte has participated in clinical trials, advisory board and/or lectures promoted by Abbvie, Almirall, Janssen, Leo Pharma, Eli Lilly, Pfizer and Sanofi-Genzyme. Maria João Paiva-Lopes has participated in clinical trials, advisory board and/or lectures promoted by AbbVie, Almirall, Boehringer Ingelheim, Janssen, Leo-Pharma, Eli Lilly, Novartis, Pfizer, Sanofi-Genzyme, Viatris. Paulo Filipe has nothing to disclose.

Ethics Approval. This study did not involve new studies with human participants or animals.

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